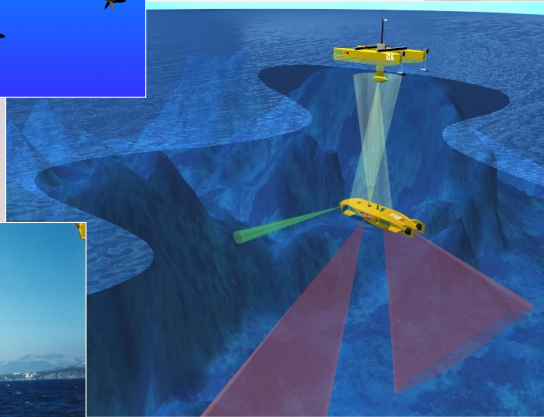
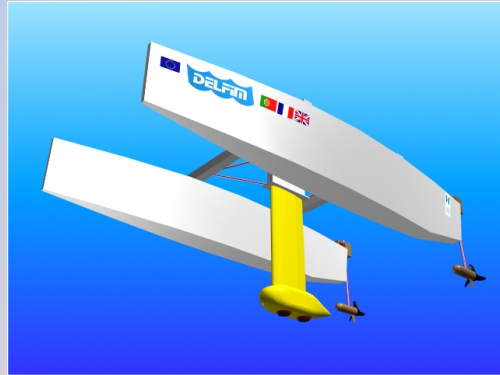
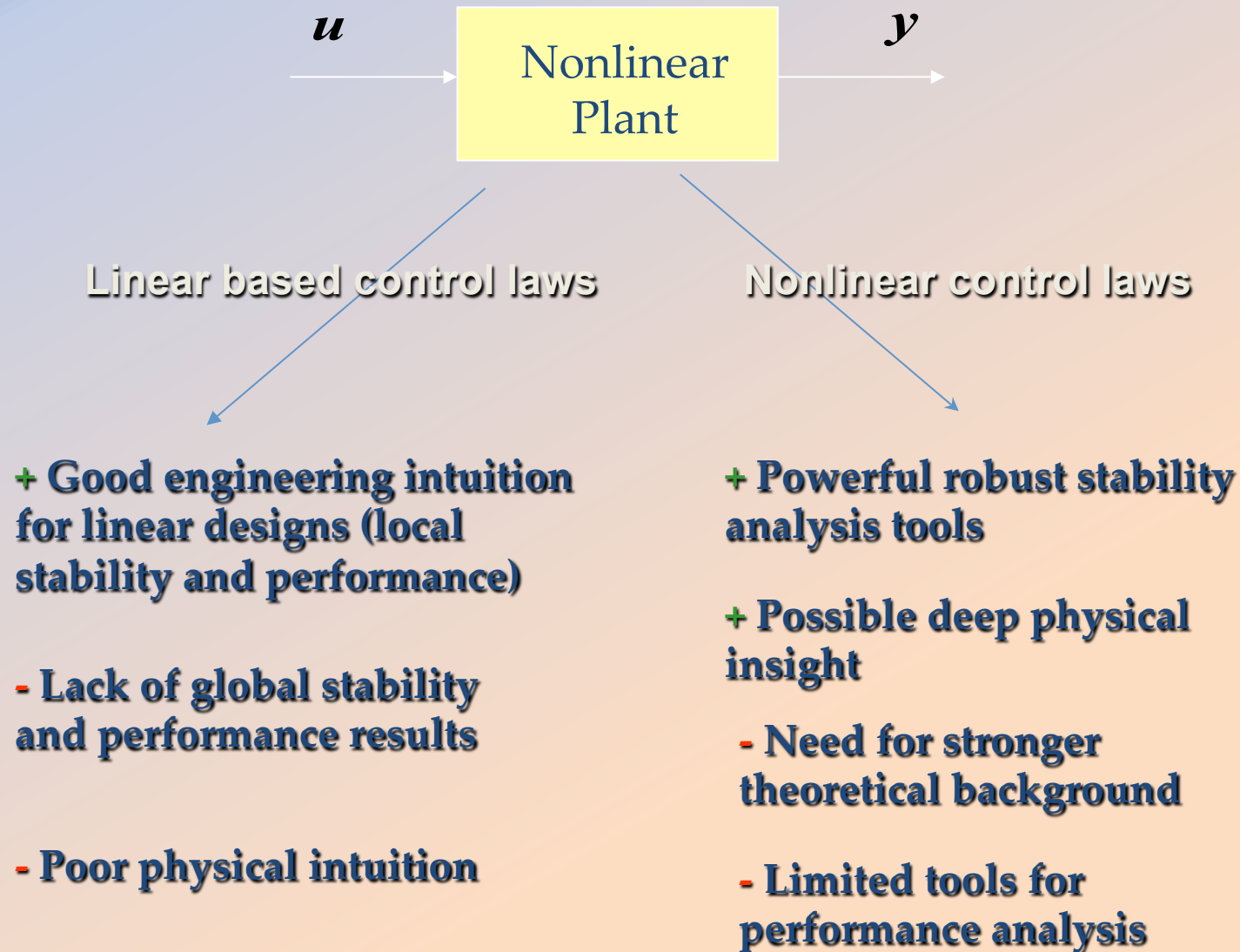


An Introduction to Lyapunov Stability Theory



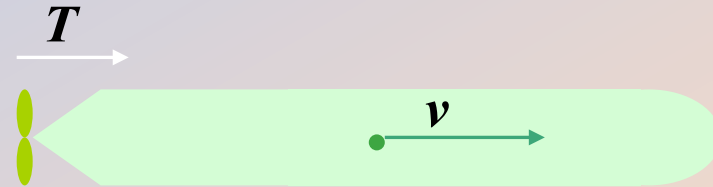
Based partially on original material by A. Pascoal

Linear versus Nonlinear Control



Nonlinear Control: Key Ingredients

AUV speed control



Dynamics

$$T - \beta v - f v^2 = m_T \frac{dv}{dt}; m_T = m + m_a$$



Objective: generate $T(t)$ so that $v(t)$ tracks the reference speed $v_r(t)$

Tracking error
$$e = v_r - v$$

Error Dynamics

$$m_T \frac{de}{dt} = m_T \frac{dv_r}{dt} - m_T \frac{dv}{dt}$$

Nonlinear Control: Key Ingredients

Error Dynamics

$$m_T \frac{de}{dt} = m_T \frac{dv_r}{dt} - m_T \frac{dv}{dt}$$

$$m_T \frac{de(t)}{dt} = m_T \frac{dv_r}{dt} - T + \beta v + f v^2$$

Nonlinear Control Law

$$T = m_T \left(\frac{dv_r}{dt} + Ke \right) + \beta v + f v^2$$

$$\longrightarrow m_T \frac{de}{dt} = m_T \cancel{\frac{dv_r}{dt}} - \left[m_T \left(\cancel{\frac{dv_r}{dt}} + ke \right) + \cancel{\beta v + f v^2} \right] + \cancel{\beta v + f v^2}$$

$$\longrightarrow \frac{de}{dt} + ke = 0 \longrightarrow e(t) = e(0) \exp(-kt); t \geq 0$$

Nonlinear Control: Key Ingredients

$$T = m_T \left(\frac{dv_r}{dt} + Ke \right) + \beta v + f v^2$$

Tracking error tends to zero exponentially fast.

$$e(t) = e(0) \exp(-kt); t \geq 0$$

Simple and elegant!

Catch: the nonlinear dynamics are known EXACTLY.

Key idea: i) use “simple” concepts, ii) deal with robustness against parameter uncertainty.

New tools are needed: LYAPUNOV theory

SIMPLE EXAMPLE

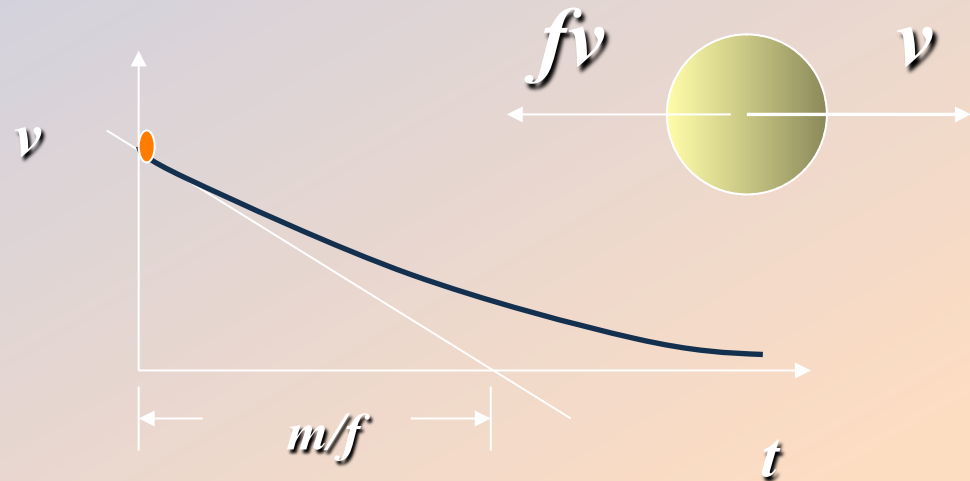
Lyapunov theory of stability: a soft Intro

(free mass, subjected to a simple motion resisting force)

$$m \frac{dv}{dt} + fv = 0$$

$$\frac{dv}{dt} = -\frac{f}{m}v$$

$$v(t) = e^{-\frac{f}{m}(t-t_0)} v(t_0)$$



$v=0$ is an equilibrium point; $dv/dt=0$ when $v=0$!

$v=0$ is attractive

***(trajectories
converge to 0)***





Lyapunov theory of stability: a soft Intro

*How can one prove that the trajectories go to the equilibrium point
WITHOUT SOLVING the differential equation?*

$$V(v) = \frac{1}{2}mv^2$$

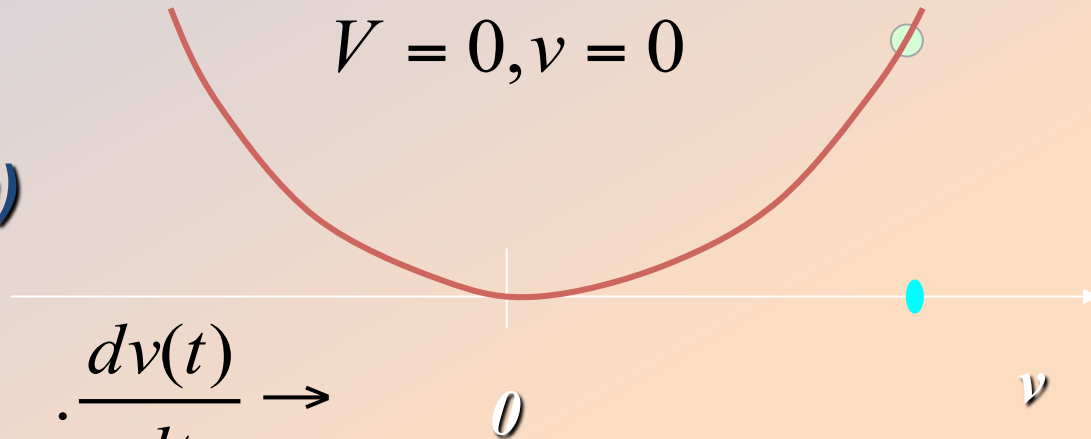
$$V(v) > 0, v \neq 0;$$

$$V = 0, v = 0$$

(energy function)

$$\frac{dV(v(t))}{dt} = \frac{\partial V}{\partial v} \bigg|_{v(t)} \cdot \frac{dv(t)}{dt} \rightarrow$$

$$\frac{dV}{dt} = mv \frac{dv}{dt} = -fv^2 < 0$$



*V positive and bounded below by zero;
dV/dt negative implies convergence
of V to 0!*



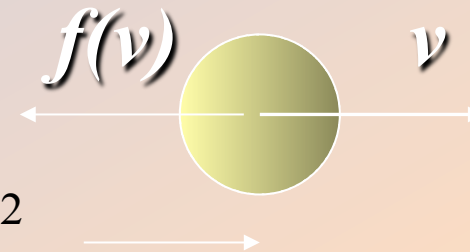
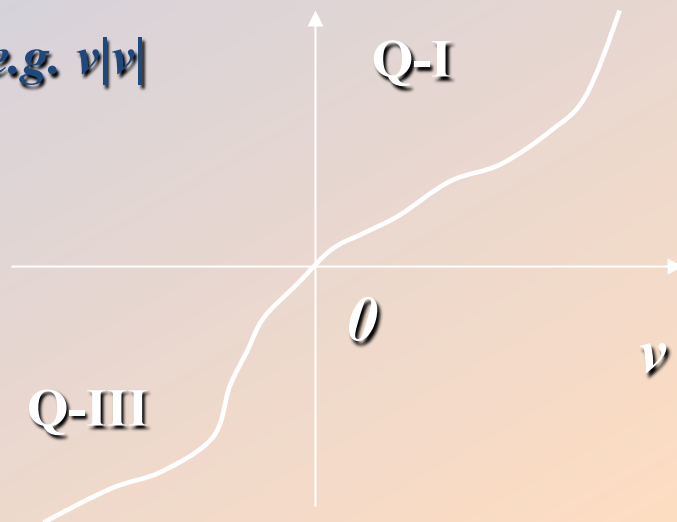
Lyapunov theory of stability: a soft Intro

*What are the **BENEFITS** of this seemingly strange approach to investigate convergence of the trajectories to an equilibrium point?*

$$m \frac{dv}{dt} + f(v) = 0$$

f a general dissipative force

e.g. $v|v|$



$$V(v) = \frac{1}{2} m v^2$$

$$\frac{dV}{dt} = m v \frac{dv}{dt} = -v f(v) < 0$$

V positive and bounded below by zero;

dV/dt negative implies convergence of V to 0!

Very general form of nonlinear equation!

2-D case

Lyapunov theory of stability: a soft Intro

$$\begin{aligned}\frac{dx_1}{dt} &= x_2 - k_1(x_1); \\ \frac{dx_2}{dt} &= -x_1 - k_2(x_2)\end{aligned}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

State vector

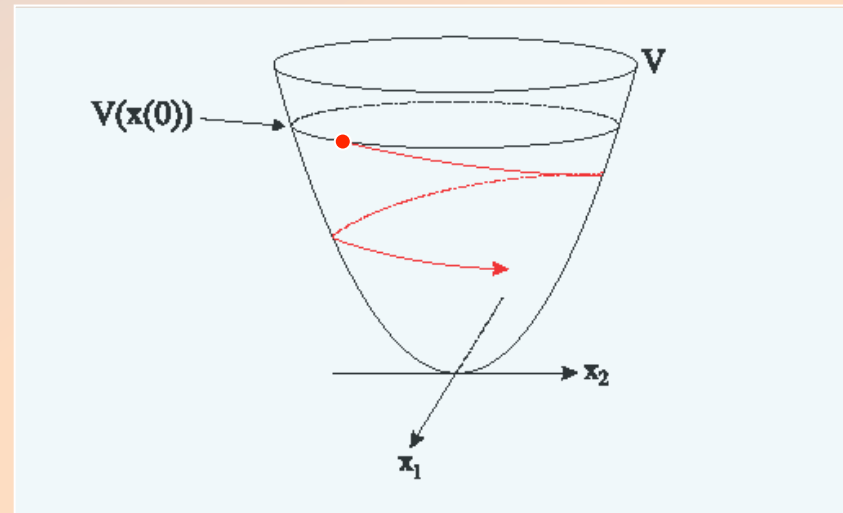
$$\frac{dx}{dt} = f(x)$$

$$\begin{aligned}V(x) &> 0, x \neq 0; \\ V &= 0, x = 0\end{aligned}$$

$$V(x) = \frac{1}{2}(x_1^2 + x_2^2)$$

$$V(x) = \frac{1}{2}x^T Qx; Q = I > 0$$

Q-positive definite



2-D case

$$x_1 \neq 0, x_1 > 0; x_2 \neq 0, x_2 > 0$$

$$V(x) = \frac{1}{2}(x_1^2 + x_2^2) \quad \begin{array}{l} \frac{dx_1}{dt} = x_2 - k_1(x_1); \\ \frac{dx_2}{dt} = -x_1 - k_2(x_2) \end{array} \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \longrightarrow \frac{dx}{dt} = f(x)$$

$$V(x(t)) \Leftarrow x(t) \Leftarrow t$$

$$V(x(t)) \in R \Leftarrow x(t) \in R^2 \Leftarrow t \in R$$

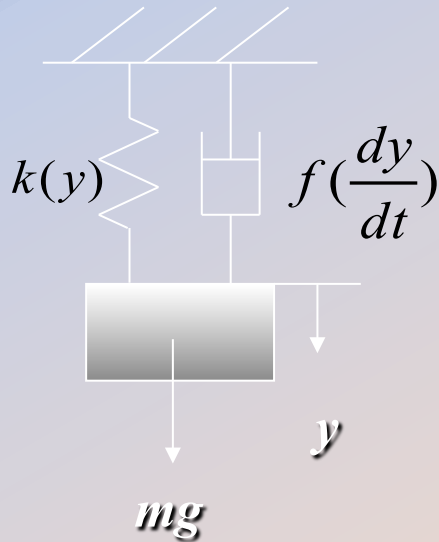
$$\underbrace{\frac{dV(x)}{dt}}_{1 \times 1} = \underbrace{\frac{\partial V}{\partial x}}_{1 \times 2} \underbrace{\frac{dx}{dt}}_{2 \times 1} \longrightarrow \frac{dV(x)}{dt} = [x_1, x_2] \begin{bmatrix} x_2 - k_1(x_1) \\ -x_1 - k_2(x_2) \end{bmatrix}$$
$$\frac{dV(x)}{dt} = x_1 x_2 - x_1 k_1(x_1) - x_2 x_1 - x_2 k_2(x_2) < 0$$

V positive and bounded below by zero;
dV/dt negative implies convergence of V to 0! $\longrightarrow$ ***x tends to 0!***

Shifting

Lyapunov theory of stability: a soft Intro

Is the origin always the **TRUE** origin?



y-measured from spring at rest

$$-k(y) - f\left(\frac{dy}{dt}\right) + mg = m \frac{d^2 y}{dt^2}$$

Equilibrium point y_{eq} : $dy/dt=0$ $k(y_{eq}) = mg$

$$x = \begin{bmatrix} y \\ \frac{dy}{dt} \end{bmatrix}; x_{eq} = \begin{bmatrix} y_{eq} \\ 0 \end{bmatrix} \quad \frac{dx}{dt} = F(x); F(x_{eq}) = 0$$

Examine if y_{eq} is “attractive”!

$$x = x_{eq} + \zeta$$

$$\frac{dx}{dt} = \frac{dx_{eq}}{dt} + \frac{d\zeta}{dt} = \frac{d\zeta}{dt}$$

$$\frac{d\zeta}{dt} = \frac{dx}{dt} = F(x_{eq} + \zeta) = G(\zeta)$$

$$G(0) = F(x_{eq} + 0) = 0$$

Examine the

ZERO eq. Point!

Shifting

Lyapunov theory of stability: a soft Intro

Is the origin always the TRUE origin?

$$\frac{dx(t)}{dt} = F(x(t)); \frac{dx_{ref}(t)}{dt} = F(x_{ref}(t))$$

$x_{ref}(t)$ is a solution



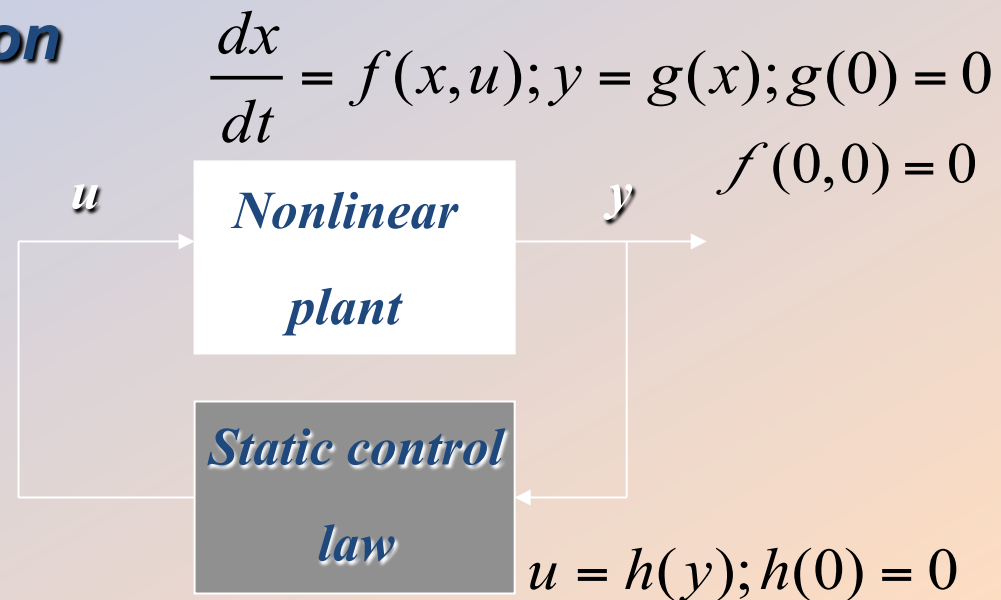
Examine if $x_{ref}(t)$ is “attractive”!

$$x = x_{ref} + \zeta$$
$$\frac{dx}{dt} = \frac{dx_{ref}}{dt} + \frac{d\zeta}{dt} =$$
$$F(x_{ref}) + \frac{d\zeta}{dt}$$
$$\frac{d\zeta}{dt} = \frac{dx}{dt} - F(x_{ref}) = F(x_{ref} + \zeta) - F(x_{ref}) = G(\zeta, t)$$
$$G(0, t) = F(x_{ref}(t) + 0) - F(x_{ref}(t)) = 0$$

*Examine the
ZERO eq. Point!*

Lyapunov theory of stability: a soft Intro

Control Action



$$\frac{dx}{dt} = f(x, h(g(x))); f(0, 0) = 0$$

$$\frac{dx}{dt} = F(x); F(0) = 0$$

Investigate if 0

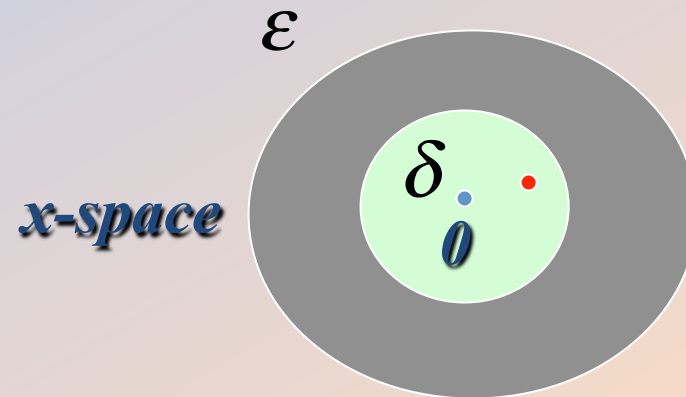
is attractive!

$$\frac{dx}{dt} = f(x) \quad ; \quad f(0) = 0$$

Lyapunov Theory

Stability of the zero solution

*The zero solution is **STABLE** if*



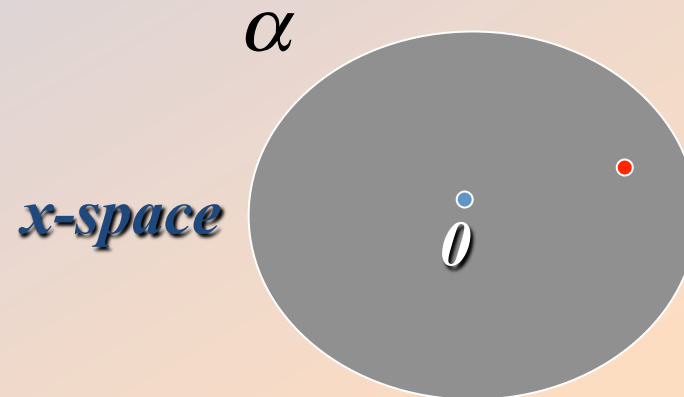
$$\forall \epsilon > 0, \exists \delta = \delta(\epsilon) > 0 : x(t_0) \in B_\delta(0) \Rightarrow x(t) \in B_\epsilon(0); \quad t \geq t_0$$

$$\frac{dx}{dt} = f(x) \quad ; \quad f(0) = 0$$

Lyapunov Theory

Attractiveness of the zero solution

The zero solution is locally ATTRACTIVE if



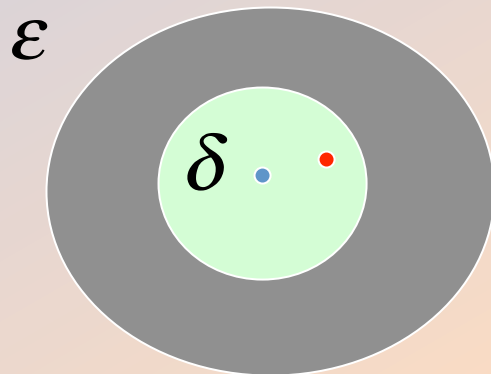
$$\exists \alpha > 0 : x(t_0) \in B_\alpha(0) \Rightarrow \lim_{t \rightarrow \infty} x(t) = 0$$

$$\frac{dx}{dt} = f(x) \quad ; \quad f(0) = 0$$

Lyapunov Theory

*The zero solution is locally
ASYMPTOTICALLY STABLE if
it is **STABLE** and **ATTRACTIVE***

**One may have attractiveness but NOT
Stability!**



**(the two conditions are required for
Asymptotic Stability!)**

Lyapunov Theory (a formal approach)

$$\frac{dx}{dt} = f(x) \quad (1)$$

Stability Definitions.

An equilibrium of the system (1) is any state $x = x_e$ which satisfies

$$f(x_e) = 0.$$

Any trajectory which takes the initial value x_e satisfies $x(t) = x_e$ for all future time. Consider an equilibrium x_e of the system (1). Without loss of generality, we assume that $x_e = 0$. (One may always make the change of coordinates $z = x - x_e$ to obtain a system with an equilibrium at the origin.)

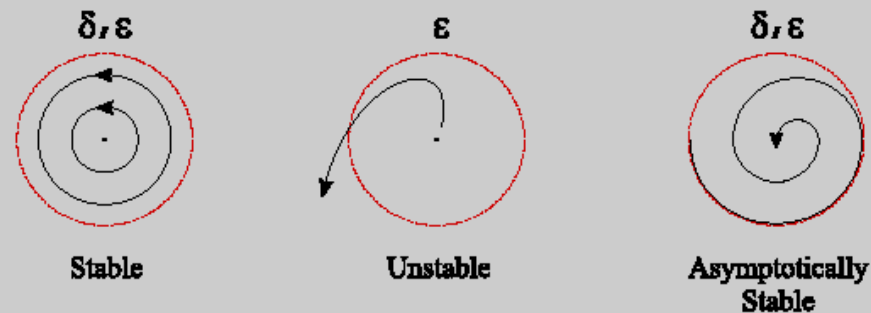


Figure 1: Stability, Instability, and Asymptotic Stability.

Lyapunov Theory

Definition: Stability. The equilibrium $\mathbf{x}_e = \mathbf{0}$ of (1) is called *stable* if for every $\epsilon > 0$, there exists a $\delta(\epsilon) > 0$ such that

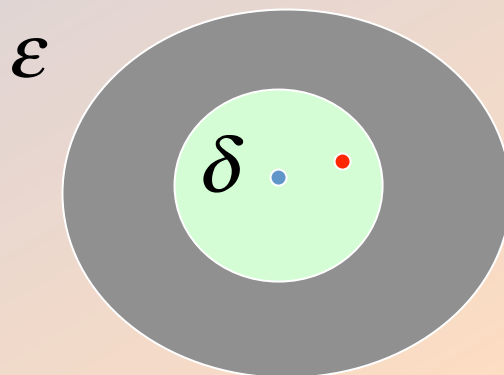
$$\|\mathbf{x}(0)\| < \delta \Rightarrow \|\mathbf{x}(t)\| < \epsilon \quad (2)$$

for every $t \geq 0$.

Definition: Instability. The equilibrium $\mathbf{x}_e = \mathbf{0}$ of (1) is called ~~*stable*~~^{unstable} if it is not stable. That is, the equilibrium is unstable if there exists an $\epsilon > 0$ for which no value of δ exists to satisfy (2).

Definition: Asymptotic Stability. The equilibrium $\mathbf{x}_e = \mathbf{0}$ of (1) is called *asymptotically stable* if it is stable and δ can be chosen such that

$$\|\mathbf{x}(0)\| < \delta \Rightarrow \lim_{t \rightarrow \infty} \mathbf{x}(t) = \mathbf{0}. \quad (3)$$



(the two conditions are required for
Asymptotic Stability!)

Lyapunov Theory

There are at least three ways of assessing the stability (of an equilibrium point of a) system:

- **Solve the differential equation (brute-force)**
- **Linearize the dynamics and examine the behaviour of the resulting linear system (local results for hyperbolic eq. points only)**
- **Use Lyapunov's direct method (elegant and powerful, may yield global results)**

Lyapunov Theory

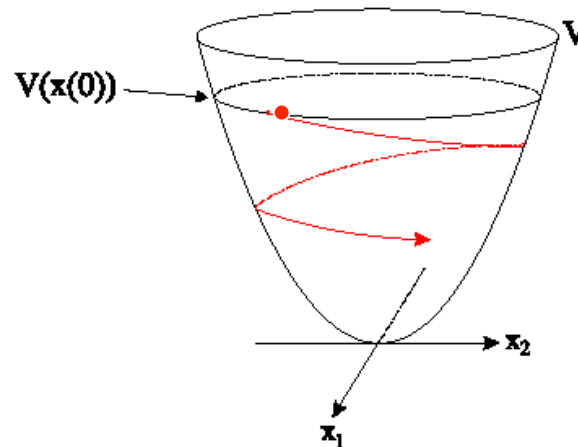
Lyapunov's Direct Method. Consider a scalar function $V(\mathbf{x})$ of the system state. This function is called *positive definite* in a domain $D \subset \mathbb{R}^n$ if

$$\begin{aligned} V(\mathbf{x}) &> 0 \quad \forall \quad \mathbf{x} \in D - \{\mathbf{0}\} \\ V(\mathbf{x}) &= 0 \quad \text{if} \quad \mathbf{x} = \mathbf{0}. \end{aligned}$$

The function is called *positive semidefinite* in a domain $D \subset \mathbb{R}^n$ if

$$V(\mathbf{x}) \geq 0 \quad \forall \mathbf{x} \in D$$

A function V is called *negative definite* (*negative semidefinite*) if $-V$ is positive definite (positive semidefinite). We will use the notation $>$, $\geq$, $<$, and $\leq$ to indicate that a function is positive definite, positive semidefinite, negative definite, or negative semidefinite, respectively.



Lyapunov Theory

Lyapunov's Stability Theorem. Let $\mathbf{x} = \mathbf{0}$ be an equilibrium point for the system

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$$

and suppose that there exists a continuously differentiable function $V(\mathbf{x})$, defined on a domain $D \subset \mathbb{R}^n$ which contains the equilibrium. If

$$\begin{aligned} V(\mathbf{x}) &> 0, \text{ and} \\ \dot{V}(\mathbf{x}) &\leq 0 \end{aligned}$$

in D , then $\mathbf{x} = \mathbf{0}$ is stable. If

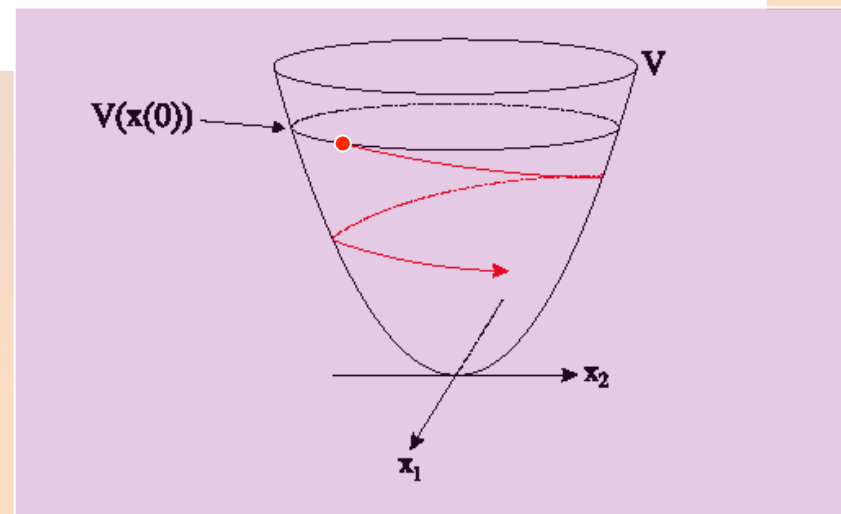
$$\begin{aligned} V(\mathbf{x}) &> 0, \text{ and} \\ \dot{V}(\mathbf{x}) &< 0 \end{aligned}$$

in D , then $\mathbf{x} = \mathbf{0}$ is asymptotically stable. $\square$

If

$$\frac{dV(x)}{dt} < 0; V(x) \rightarrow \infty \text{ as } \|\mathbf{x}\| \rightarrow \infty$$

**then the origin is
globally asymptotically
stable**



Lyapunov Theory

What happens when $\frac{dV(x)}{dt} \leq 0$?

Is the situation hopeless? **No!**

Let $\frac{dV(x)}{dt} \leq 0$; define

$$\Omega := \left\{ x : \frac{dV(x)}{dt} = 0 \right\}$$

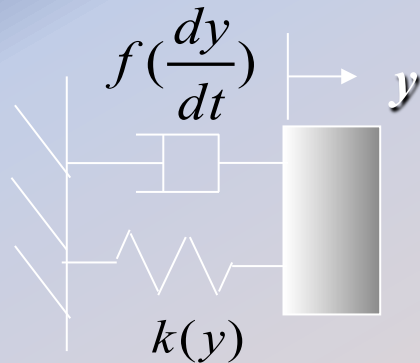
Suppose the only trajectory of the system entirely contained in Ω is the null trajectory. Then, the origin is asymptotically stable

Krazovskii-LaSalle

(Let M be the largest invariant set contained in Ω . Then all solutions converge to M . If M is the origin, the results follows)

Krazovskii-La Salle

Lyapunov Theory



$$-k(y) - f\left(\frac{dy}{dt}\right) = m \frac{d^2 y}{dt^2}$$

$$x = \begin{bmatrix} y \\ \frac{dy}{dt} \end{bmatrix}; x_{eq} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

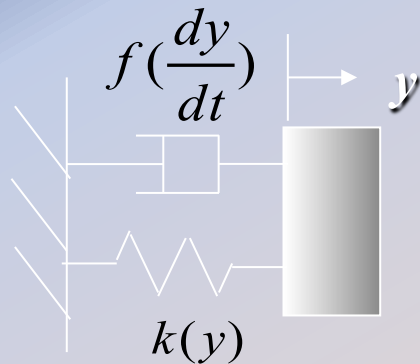
$$\begin{bmatrix} \frac{dx_1}{dt} \\ \frac{dx_2}{dt} \end{bmatrix} = \begin{bmatrix} x_2 \\ -\frac{1}{m} k(x_1) - \frac{1}{m} f(x_2) \end{bmatrix} \longrightarrow \frac{dx}{dt} = F(x); F(0) = 0$$

$$V(x) = \text{Kinetic Energy} + \text{Potential Energy}$$

$$V(x) = \frac{1}{2} m x_2^2 + \int_0^{x_1} k(\zeta) d\zeta$$

Krazovskii-La Salle

Lyapunov Theory



$f(\cdot), k(\cdot)$ – 1st and 3rd quadrants

$$f(0)=k(0)=0$$

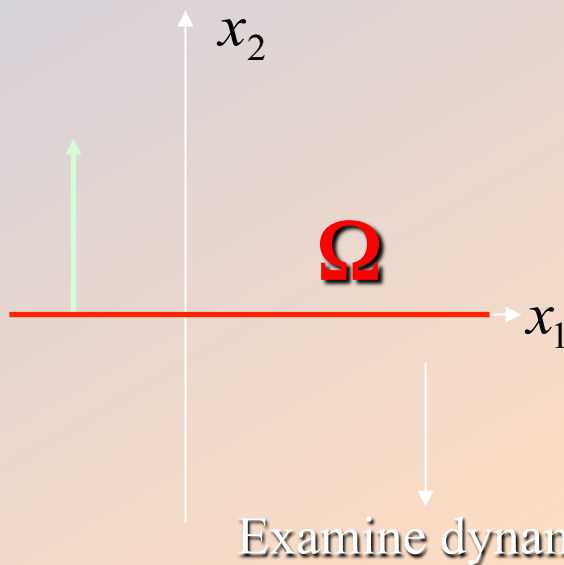
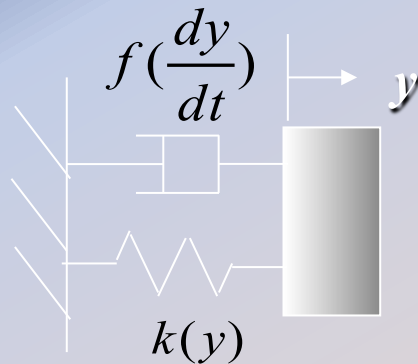
$$\begin{bmatrix} \frac{dx_1}{dt} \\ \frac{dx_2}{dt} \end{bmatrix} = \begin{bmatrix} x_2 \\ -\frac{1}{m}k(x_1) - \frac{1}{m}f(x_2) \end{bmatrix} \quad V(x) = \frac{1}{2}mx_2^2 + \int_0^{x_1} k(\zeta)d\zeta$$

$$V(x) > 0!$$

$$\frac{dV(x)}{dt} = mx_2 \frac{dx_2}{dt} + k(x_1) \frac{dx_1}{dt} =$$

$$mx_2 \left(-\frac{1}{m}k(x_1) - \frac{1}{m}f(x_2) \right) + k(x_1)x_2 = -f(x_2)x_2 \leq 0!$$

Krazovskii-La Salle



Lyapunov Theory

$$\begin{bmatrix} \frac{dx_1}{dt} \\ \frac{dx_2}{dt} \end{bmatrix} = \begin{bmatrix} x_2 \\ -\frac{1}{m}k(x_1) - \frac{1}{m}f(x_2) \end{bmatrix}$$

$$\frac{dV(x)}{dt} - f(x_2)x_2 \leq 0!$$

$$\frac{dV}{dt} = 0 \text{ for } x_2 = 0!$$

$$\begin{bmatrix} \frac{dx_1}{dt} \\ \frac{dx_2}{dt} \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{1}{m}k(x_1) \end{bmatrix}$$

***Trajectory
leaves Ω
unless $x_1=0$!***

M is the origin.

***The origin is
asymptotically
stable!***

An application example (physically motivated control law)

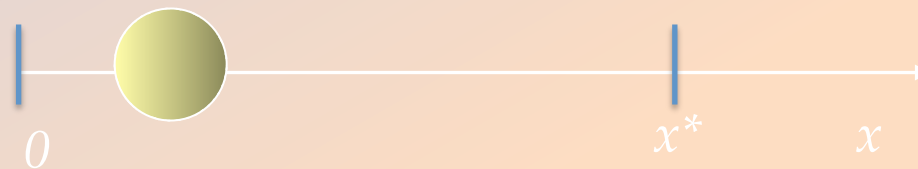
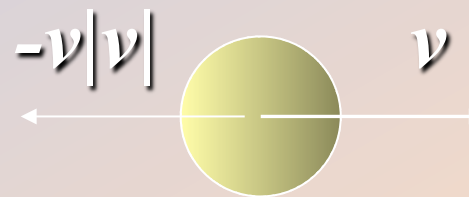
Krazovskii-La Salle

Lyapunov Theory

AUV moving in the water with speed v
under the action of the applied force u .

$$\frac{dv}{dt} = -v |v| + u$$

$$\frac{dx}{dt} = v$$



Objective: drive the position x of the AUV to x^* (by proper choice of u)

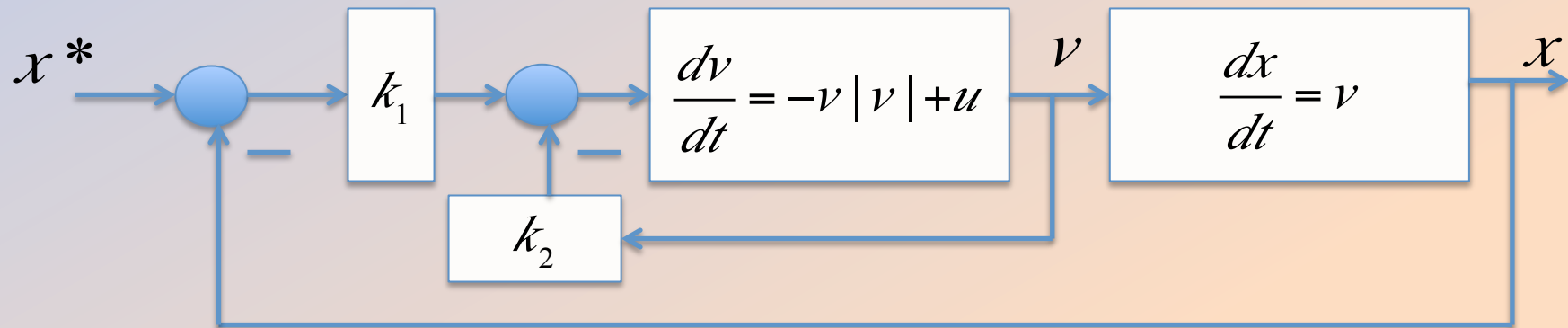
Krazovskii-La Salle *An application example* Lyapunov Theory

Suggested control law

P-D control

$$u = -k_1(x - x^*) - k_2 v; \quad k_1, k_2 > 0$$

$$\frac{dv}{dt} = -v |v| + u$$
$$\frac{dx}{dt} = v$$



Control law exhibits Proporcional + Derivative actions

The plant itself has a pure integrator (to drive the static error to 0)

Krazovskii-La Salle Lyapunov Theory

Show asymptotic stability of the (equilibrium point of the) system

Step 1. Start by re-writing the equations in terms of the variables that must be driven to 0.

Objective:

$$v(t) \rightarrow 0$$

$$e(t) = x(t) - x^*(t) \rightarrow 0$$

$$\frac{dv}{dt} = -v|v| + u$$



$$\frac{de}{dt} = \frac{dx}{dt} - \frac{dx^*}{dt} = \frac{dx}{dt} = v$$

$$u = -k_1(x - x^*) - k_2v; \quad k_1, k_2 > 0$$



$$\begin{aligned} \frac{dv}{dt} &= -v|v| - k_1e - k_2v \\ \frac{de}{dt} &= v \end{aligned}$$

Krazovskii-La Salle Lyapunov Theory

Show asymptotic stability of the (equilibrium point of the) system

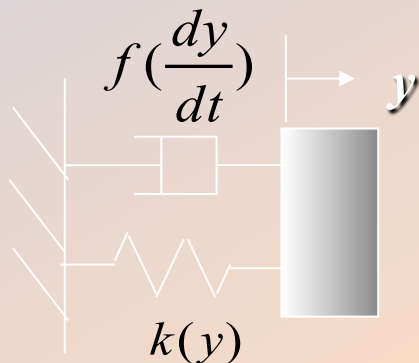
$$\frac{dv}{dt} = -v|v| - k_1 e - k_2 v \quad [v, e]^T = [0, 0]^T$$

$$\frac{de}{dt} = v$$

is an equilibrium point of the system!

Step 2. Prove global asymptotic stability of the origin

Seek inspiration from the spring-mass-dashpot system



$V(x) = \text{Kinetic Energy} + \text{Potential Energy}$

$$V(x) = \frac{1}{2} v^2 + \int_0^e k_1 \varsigma d\varsigma$$

Krazovskii-La Salle Lyapunov Theory

**Show asymptotic stability of the
(equilibrium point of the) system**

$$V(x) = \text{Kinetic Energy} + \text{Potential Energy} \quad V(x) = \frac{1}{2}v^2 + \int_0^e k_1 \zeta d\zeta$$

$$V(x) > 0!$$

$$\begin{aligned} \frac{dV}{dt} &= \frac{\partial V}{\partial v} \frac{dv}{dt} + \frac{\partial V}{\partial e} \frac{de}{dt} \\ &= v \frac{dv}{dt} + k_1 e \frac{de}{dt} = v[-v|v| - k_1 e - k_2 v] + k_1 ev \end{aligned}$$

$$= -v^2|v| - k_2 v^2 \leq 0!$$

**Using La Salle's theorem it follows that the origin is
globally asymptotically stable (GAS)**

La Salle Example

Example- Damped pendulum

$$\ddot{\theta} + b\dot{\theta} + \sin \theta = 0 \Rightarrow \begin{pmatrix} \dot{\theta} \\ \dot{\omega} \end{pmatrix} = \begin{pmatrix} \omega \\ -\sin \theta - b\omega \end{pmatrix}$$

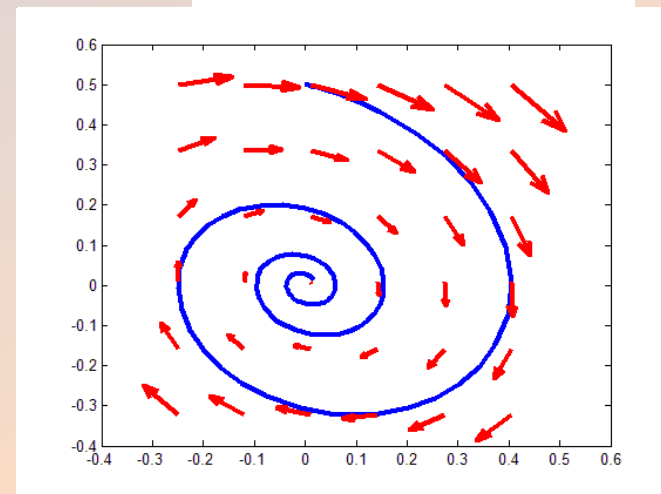
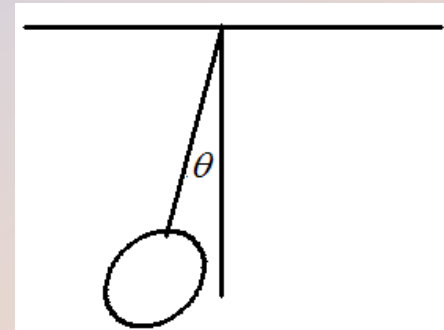
(1) Propose

$$V(\theta, \omega) = (1 - \cos \theta) + \frac{1}{2} \omega^2$$

(2) Derivative

$$\begin{aligned} \dot{V}(\theta, \omega) &= \sin \theta \cdot \omega + \omega \dot{\omega} \\ &= -b\omega^2 \leq 0 \end{aligned}$$

$\Rightarrow$ Asymptotic Stability of Origin



Construction for Linear Systems

For a Linear System:

$$\dot{x} = Ax$$

(1) Propose

$$V(x) = x^T Px$$

P is symmetric

P is positive definite

(2) Time Derivative

$$\dot{V}(x) = \nabla V \cdot Ax = x^T (PA + A^T P)x$$

If we choose $Q \leq 0$ and solve algebraically for P:

$$PA + A^T P = Q$$

As long as A is stable, a solution is known to exist.

Also an explicit representation of the solution exists:

$$P = \int_0^{\infty} e^{A^T t} Q e^{At} dt$$

General Construction Systems

How to choose the Lyapunov Function ?

Construction methods for an Arbitrary System $\dot{x} = f(x)$

Krasofskii

$V(x) = f^T P f$ A quadratic form (ellipsoid) of system velocity

$$\dot{V}(x) = \dot{f}^T P f + f^T P \dot{f} = f^T \left(\frac{\partial f^T}{\partial x} * P + P * \frac{\partial f}{\partial x} \right) f$$

Solve

$$\frac{\partial f^T}{\partial x} * P + P * \frac{\partial f}{\partial x} = Q \leq 0$$

Variable Gradient

$$V(x) = \int_0^x \nabla V dx$$

Assume a form for the gradient, i.e. $\nabla V_i = \sum_{j=1}^n a_{ij} x_j$ [Slotine, Li]
Solve for negative semi-definite gradient
Integrate and hope for positive definite V [Hahn]

EXAMPLES:

Example 1: Consider the system

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -x_1 + \frac{1}{16}x_1^5 - x_2$$

Calculate the equilibrium points for the system.

Plot the phase portrait of the system.

Solution:

The equilibrium points must be stationary. Therefore for the first system we have

$$0 = x_2$$

$$0 = x_2$$

$$0 = -x_1 + \frac{1}{16}x_1^5 - x_2$$

$$0 = -x_1 + \frac{1}{16}x_1^5 = -x_1(1 - 0.0625x_1^4)$$

$$0 = x_2$$

$$0 = -x_1 + \frac{1}{16} x_1^5 = -x_1 \left(1 - 0.0625 x_1^4 \right)$$

$$x_1 = 0$$

$$\text{roots}([-1/16 \ 0 \ 0 \ 0 \ 1])$$

ans =

$$-2.0000$$

$$-0.0000 + 2.0000i$$

$$-0.0000 - 2.0000i$$

$$2.0000$$

The Jacobian matrix is defined as

$$J = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 + \frac{5}{16} x_1^4 & -1 \end{bmatrix}$$

The equilibrium points are

$$x_e = [(0,0), (2,0), (-2,0)]$$

$$J_{x_{e1}=(0,0)} = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$$

$\text{eig}(J) \Rightarrow$

$$-0.5000 + 0.8660i$$

$$-0.5000 - 0.8660i$$

Stable node

```
[x1, x2] = meshgrid(-4:0.2:4, -2:0.2:2);
```

```
x1dot = x2;
```

```
x2dot = -x1+(1/16)*x1.^5-x2;
```

```
quiver(x1,x2,x1dot,x2dot)
```

```
xlabel('x_1')
```

```
ylabel('x_2')
```

$$J_{x_{e2}=(-2,0)} = \begin{bmatrix} 0 & 1 \\ -1 + \frac{5}{16}(-2)^4 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 4 & -1 \end{bmatrix}$$

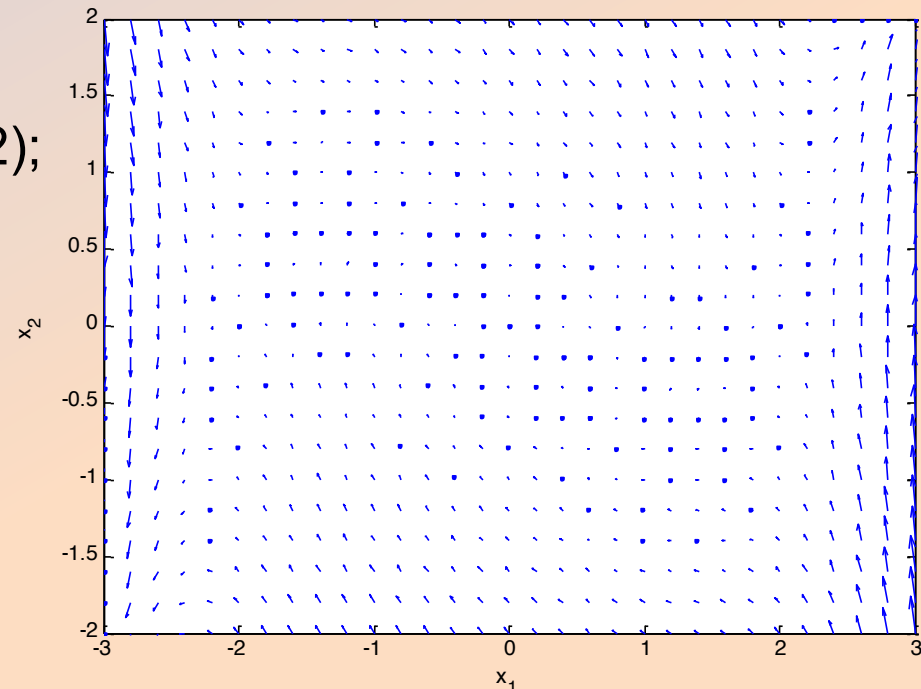
$\text{eig}(J) \Rightarrow$

$$1.5616$$

$$-2.5616$$

The same result is
obtained for $x_{e3} (2,0)$

Saddle points



Example 2. Show that the origin of the system is stable, using a suitable Lyapunov function.

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -x_1^3 - x_2^3$$

Solution: Let us use the following Lyapunov function

$$V(x) = \frac{1}{4} x_1^4 + \frac{1}{2} x_2^2$$

$$\begin{aligned}\dot{V}(x) &= \frac{dV}{dt} = \frac{\partial V}{\partial x} \frac{dx}{dt} = \nabla V \cdot f(x) \\ &= \left[\frac{\partial V}{\partial x_1}, \frac{\partial V}{\partial x_2} \right] \begin{bmatrix} f_1(x) \\ f_2(x) \end{bmatrix}\end{aligned}$$

$$\begin{aligned}\dot{V}(x) &= \begin{bmatrix} x_1^3 & x_2 \end{bmatrix} \begin{bmatrix} x_2 \\ -x_1^3 - x_2^3 \end{bmatrix} \\ \dot{V}(x) &= x_1^3 x_2 + x_2 (-x_1^3 - x_2^3) \\ \dot{V}(x) &= -x_2^4 \leq 0\end{aligned}$$

The system is stable in the sense of Lyapunov.

LYAPUNOV STABILITY FOR LINEAR TIME-INVARIANT SYSTEMS:

Given a linear system of the form

$$\dot{\mathbf{x}} = \mathbf{A} \mathbf{x}$$

Let us consider a quadratic Lyapunov function candidate

$$V = \mathbf{x}^T \mathbf{P} \mathbf{x}$$

where $\mathbf{P}$ is a given symmetric positive definite matrix.

Positive definite matrices

Given a symmetric two by two matrix $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$, here are four ways to tell if it's positive definite:

1. Eigenvalue test: $\lambda_1 > 0, \lambda_2 > 0$.
2. Determinants test: $a > 0, ac - b^2 > 0$.
3. Pivot test: $a > 0, \frac{ac - b^2}{a} > 0$.
4. $\mathbf{x}^T \mathbf{A} \mathbf{x}$ is positive except when $\mathbf{x} = \mathbf{0}$ (this is usually the definition of positive definiteness).

Differentiating the positive definite function V along the system trajectory yields another quadratic form

$$\dot{V} = \dot{x}^T P x + x^T P \dot{x}$$

where

$$\begin{aligned}\dot{V} &= (A x)^T P x + x^T P (A x) \\ &= x^T A^T P x + x^T P A x \\ &= x^T (A^T P + P A) x \quad \leftarrow \quad A^T P + P A = -Q \\ &= -x^T Q x\end{aligned}$$

If there exists a positive definite matrix Q satisfying the equation (Lyapunov equation), the system is said to be stable in the sense of Lyapunov (ISL).

$$A^T P + P A + Q = 0 \quad \text{Lyapunov equation.}$$

Example:

Consider two matrices,

$$A = \begin{bmatrix} 0 & 1 \\ -12 & -8 \end{bmatrix}, Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The linear system is stable (Real parts of all eigenvalues of the system matrix A are negative) if there is a positive definite matrix P.

Using Matlab, we can find the matrix P as

```
clc;clear;  
A=[0 1;-12 -8];  
Q=[1 0;0 1];  
P=lyap(A,Q)  
eig(P)
```

The matrix P is positive definite, since the eigenvalues are real, and the system is stable ISL.

```
P =  
0.4010 -0.5000  
-0.5000 0.8125  
  
ans =  
0.0661  
1.1474
```

>0

LYAPUNOV FUNCTION FOR NONLINEAR SYSTEM:

Krasovskii's method suggests a simple form of Lyapunov function candidate (LFC) for autonomous nonlinear systems, namely, $V=f^T f$. The basic idea of the method is simply to check whether this particular choice indeed leads to a Lyapunov function.

Theorem (Krasovskii): Consider the autonomous system defined by $\frac{dx}{dt}=f(x)$, with the equilibrium point of interest being the origin. Let $J(x)$ denote the Jacobian matrix of the system, i.e.,

$$J(x) = \frac{\partial f}{\partial x}$$

If the matrix $F=J+J^T$ is negative definite, the equilibrium point at the origin is asymptotically stable. A Lyapunov function for this system is

$$V(x) = f^T(x)f(x)$$

If $V(x) \rightarrow \infty$ as $\|x\| \rightarrow \infty$, then the equilibrium point is globally asymptotically stable.

Example:

Consider a nonlinear system

$$\dot{x}_1 = -6x_1 + 2x_2$$

$$\dot{x}_2 = 2x_1 - 6x_2 - 2x_2^3$$

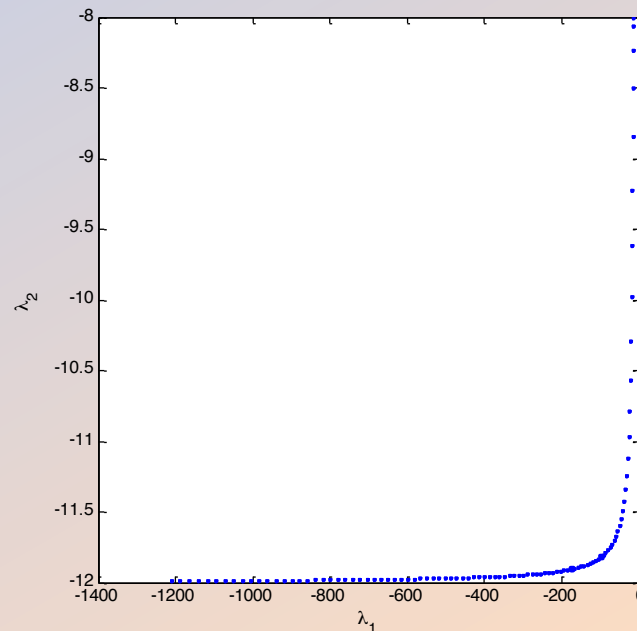
We have

$$J = \frac{\partial f}{\partial x} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} -6 & 2 \\ 2 & -6 - 6x_2^2 \end{bmatrix}$$

$$F = J + J^T = \begin{bmatrix} -12 & 4 \\ 4 & -12 - 12x_2^2 \end{bmatrix}$$

The matrix F is negative definite over the whole state space. Therefore, the origin is asymptotically stable, and a Lyapunov function candidate is

```
clc;clear;
x2=-10:0.1:10;
for i=1:length(x2)
F=[-12 4;4 -12-12*x2(i)^2];
eg=eig(F)
plot(eg(1),eg(2))
hold on
end
```



$$V(x) = f^T(x)f(x) = \begin{bmatrix} (-6x_1 + 2x_2) \\ (2x_1 - 6x_2 - 2x_2^3) \end{bmatrix} \begin{bmatrix} (-6x_1 + 2x_2) \\ (2x_1 - 6x_2 - 2x_2^3) \end{bmatrix}$$

$$V(x) = f^T(x)f(x) = (-6x_1 + 2x_2)^2 + (2x_1 - 6x_2 - 2x_2^3)^2$$

Since $V(x) \rightarrow \infty$ as $\|x\| \rightarrow \infty$, then the equilibrium point is globally asymptotically stable.

Example (Variable Gradient Method):

Consider a nonlinear system

$$\dot{x}_1 = -2x_1$$

$$\dot{x}_2 = -2x_2 + 2x_1x_2^2$$

We assume that the gradient of the undetermined Lyapunov function has the following form

$$\nabla V_1 = a_{11}x_1 + a_{12}x_2$$

$$\nabla V_2 = a_{21}x_1 + a_{22}x_2$$

The curl equation is

$$\frac{\partial \nabla V_1}{\partial x_2} = \frac{\partial \nabla V_2}{\partial x_1}$$

$$a_{12} + x_2 \frac{\partial a_{12}}{\partial x_2} = a_{21} + x_1 \frac{\partial a_{21}}{\partial x_1}$$

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If the coefficients are chosen to be

$$a_{11}=a_{22}=1, a_{12}=a_{21}=0$$

which leads to

$$\nabla V_1 = x_1 \quad \nabla V_2 = x_2$$

Then

$$\dot{V} = \nabla V \dot{x} = -2x_1^2 - 2x_2^2(1 - x_1x_2)$$

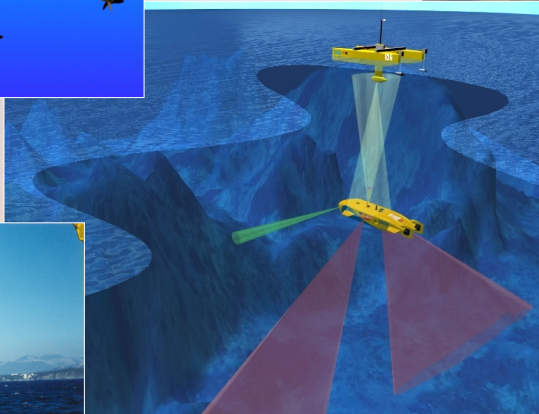
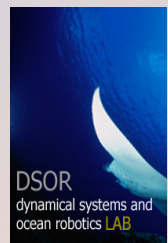
Thus, dV/dt is locally negative definite in the region $(1-x_1x_2)>0$. the function V can be computed as

$$V(x) = \int_0^{x_1} x_1 dx_1 + \int_0^{x_2} x_2 dx_2 = \frac{x_1^2 + x_2^2}{2}$$

This is indeed positive definite, and therefore the asymptotic stability is guaranteed.

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An Introduction to Lyapunov Stability Theory



Based on the original material
by A. Pascoal